5. A potential difference of 100 V exists across two parallel plates that are 16.0 cm apart. An electron begins at rest at one of the plates and accelerates toward the second plate. How fast is the electron moving when it has traveled 4.00 cm? How fast is it moving when it has traveled 8.00 cm? How fast is it moving when it has traveled 16.0 cm?

6. An electron begins at a speed of 5.00×10^6 m/s and is slowed down by a potential difference of 15.0 V. What is the final speed of the electron?

7. An electron enters the gap between parallel plates (potential difference of 20.0 V, 10.0 cm apart) at 6.00×10^6 m/s. How fast is the electron moving when it reaches the second plate if it is moving from the negative plate to the positive plate? How fast is the electron moving when it reaches the second plate if it is moving from the positive plate to the negative plate?